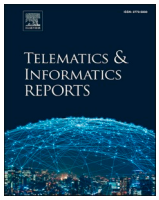
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Blockchain technology in maritime single window and port community systems: A bibliometric analysis and systematic literature review

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ABSTRACT

This paper presents systematic literature review with bibliometric analysis that aims at studying and identifying potential uses of blockchain technology in the context of Maritime Single Window and Port Community Systems. The research aims to examine the state of development of the blockchain technology, to determine the trends of the industry and to analyze the prevalent strengths and weaknesses of the industry. Through analyzing 51 peer-reviewed articles from Scopus and Web of Science (between 2013 and 2023), it is possible to state that the rate of scholars' engaging with the topic has risen significantly since 2018 within the countries in Asia, especially China, Singapore, and other countries. Our mapping outlines five major streams of research: security and sharing of data in ports and port innovation; behavioral analysis of decentralized technologies in smart ports; factors leading to success of blockchains; effects of blockchains on container shipping; and integration of blockchains with IoT and supply chain management. The review highlights how blockchain can improve security, increase the performance of procedures, guarantee signature and credibility, and build trust, which are fundamental for growth in the maritime sectors. However, the study identifies several main implementation challenges, namely, scale-up features, interoperability, open-endedness, institutional implications, and variable management. As a result, we offer suggestions to the practitioners, policymakers, and scholars on how best to go about the adoption and implementation of blockchain in the maritime industry. Further research in the form of implementation studies, case, legal, and probes on how Blockchain interacts with other emerging technologies is recommended in the study. Namely, they establish foundational domains and define further methodology for subsequent studies and advancement related to blockchain enablement of maritime industries.

1. Introduction

Maritime business is on the brink of a digital transformation. Blockchain technology is seen as a key driver for change in the Maritime Single Window (MSW) and Port Community Systems (PCS) context. Maritime Single Window functions as an electronic system which permits ships and port authorities to swap standardized information based on International Maritime Organization (IMO) rules for digital vessel call submissions [1]. The system mainly enables B2G communication to handle administrative formalities. A neutral open electronic platform named PCS enables all public and private actors within the port community to conduct secure and intelligent information management [2]. The system focuses on Business-to-Business (B2B) exchanges while simultaneously optimizing operational and logistics processes that take place in the port. MSW and PCS are digital platforms that support the communication of supply chain participants, including port authorities, customs, shipping companies, and freight forwarders [3,4]. Such systems can be improved by using blockchain technology since it can help simplify various processes, minimize the use of papers, improve data quality and credibility, and increase participant trust [5]. Blockchain is a distributed ledger technology that tends to register transactions with secure transparency beyond central control [6]. The system operates differently than regular databases since it creates features which combine data immutability with decentralization and distributed consensus mechanisms [7]. The maritime sector benefits more from blockchain technology compared to EDI and conventional database systems because blockchain enhances security along with data transparency and implements automatic operations using smart contracts [8]. Currently, permissioned blockchain solutions in the context of the shipping industry range from the Global Shipping Business Network based on Corda to the CargoX platform based on Ethereum as well as public blockchains developed based on Ethereum. The choices also

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differ in terms of consensus mechanisms, scalability, and the system of governance. Private blockchains allow for the additional privacy, though it is at the expense of decentralization as compared to permissionless public blockchains. In implementing blockchain for managing MSW and PCS, there are five core processes that must be included: (a) the consolidation & e-validation of regulatory documents, including manifests and customs declaration; (b) the strategic planning for berthing operations; (c) the negotiation and coordination of execution between stakeholders; (d) the identification and processing of commercial transactions; and (e) the real-time sharing of data. These processes altogether improve the efficiency and effectiveness of the overall structures in maritime logistics. [9,10]. Despite the potential of blockchain technology in the maritime industry, particularly in the aspects of MSW and PCS, its application has not yet been widely seen. It may be useful to consider the related literature to determine the possible measures to support the integration of this technology. Hence, from the literature analysis of blockchain applications in the context of MSW and PCS, it is possible to identify the benefits of the technology, the issues, and the challenges that may hamper its implementation. This paper offers a bibliometric and systematic literature review of the use of blockchain technology in MSW and PCS. This work is focused on analyzing the current state of knowledge, the defining themes and trends, and the advantages and challenges of blockchain application in this field. Maritime business is plagued by problems like document management, lack of real-time data, and low-level trust between the parties involved that may result in slow processes, errors, and added expenses [9,11]. These challenges can be addressed by leveraging blockchain technology, which offers security, efficient data sharing, enables smart contract implementation, and facilitates the involvement of all parties [12,13]. Blockchain serves as a specialized tool that addresses distributed transparent and immutable records but it does not provide remedies for every maritime issue [14]. The application process must evaluate blockchain technology using specific benchmarks including distributed trust requirements together with process complexity measures and actor network structures [15]. Further research on the factors that enable and hinder the contextual escalation of blockchain in both MSW and PCS from the technological readiness perspective, along with the necessary regulatory enactments and stakeholders' acceptance, could provide sufficient knowledge to policy makers and practitioners. Reviews of the literature have also revealed recent efforts in the maritime environment to extend the use of blockchain in other areas like supply chain management, cargo, and trade finance [16]. Thus, the application of blockchain has been suggested in areas such as managing the bill of lading, customs, and tracking of containers, which results in minimizing paperwork, improving the effectiveness of the supply chain, and increasing transparency [17,18]. Blockchain applications for MSW and PCS should provide maritime solutions that deal with the multiple port information system fragmentation and lack of trust between actors at sea and paper-based process reliance and bill of lading security concerns [19,20]. The market has become more aware of blockchain technology by seeing evolution like the Global Shipping Business Network and the TradeLens platform, initiated in 2018 till 2023 operated by Maersk-IBM partnership. Nevertheless, the actual implementation depends on the further advancement [21]. Despite blockchain's benefits to the maritime industry presently, there is still much work to be done to integrate the technology; this is because some challenges still need to be addressed. These include a lack of regulations governing blockchain use, compatibility issues between the different blockchain platforms, and the need to standardize the technology [11]. Thus, the applicability of Blockchain in MSW and PCS depends on stakeholders' participation, standardization of the industry, and consistent legal and regulatory frameworks [22,23]. This paper enriches the existing literature on the use of blockchain technology in the maritime sector by concentrating on two applications, namely MSW and PCS. Our research takes a different approach than earlier maritime blockchain reviews because it concentrates on MSW and PCS rather than general maritime supply chain studies [16] and merges bibliometric and systematic strategies and establishes five research clusters as a structured field taxonomy. Based on the literature analysis, the thematic and trend analysis, as well as the discussion of opportunities and risks of blockchain implementation in this area, this paper can be helpful for researchers, practitioners, and policymakers who are interested in applying the blockchain concept to improve the efficiency and performance of the maritime sector [11,18]. The remainder of the paper is organized as follows. Section 2 provides an overview of the research methodology employed in this study, including the bibliometric analysis and systematic literature review. The findings of the bibliometric analysis is presented in detail in Section 3, including trends, authors, and journals. Section 4 describes five main research clusters from the systematic review, discusses the opportunities and barriers of applying Blockchain for MSW and PCS, and provides research recommendations. In the last section of the paper, the conclusions, the relationships between the findings of the present research and the discussed issues, and suggestions for future actions for the maritime industry and their governing bodies will be provided.

2. Methodology

This section discusses the research methodology using both the bibliometric analysis and the systematic literature review (SLR) by defining the Research Design, Database, Selection and Search Strategy, Inclusion and Exclusion Criteria, and Data Extraction and Synthesis.

2.1. Research design

This study employed bibliometric analysis and systematic review to achieve its research objectives. Bibliometric analysis, a common literature review technique in business research, offers an overview of the development of the use of blockchain technology in Maritime Single Window (MSW) and Port Community Systems (PCS) [24]. This method provides a structured, clear, and replicable approach by clearly and effectively mapping quantitative data using VOSviewer as a tool [25]. Furthermore, to gain profound insights into current research interests, an exploratory systematic review was conducted, adhering to a three-step process: setting criteria for inclusion and exclusion of studies, categorizing the papers, and interpreting the results in light of the identified criteria [26]. The use of a systematic review is especially warranted in the case of a research domain that is still in the early development stage [27].

2.2. Database selection and search strategy

In this academic study, we considered only two databases since very relevant peer-reviewed article: Scopus and Web of Science databases were selected to conduct the literature review. These repositories were selected because they indexed a huge number of peer-reviewed articles in various scholarly journals, as well as being well-known among the academic audience. Consequently, the collection of data for this study required the formulation of specific search strings aimed in these databases to obtain articles relevant to the research. In the case of Scopus and Web of Science, the search string used was (TITLE/ABS/KEY) (blockchain or smart contract or ledger) AND (port community system or single window or maritime or port). These queries were very carefully constructed in order to aim at finding articles that specifically discuss the use of blockchain in the context of Maritime Single Windows (MSW) and Port Community Systems (PCS). This research employed a research design that sought to find all blockchain applications for maritime purposes more so where MSW and PCS operations were involved. To ensure that all the latest technologies and potential maritime applicability related to blockchain was captured, "smart contract" and "ledger" were added to the search terms because these could be synonymous with blockchain in some cases [28]. Moreover, the terms 'maritime' and

'port' were used along with 'port community system' and 'single window' terms to include researches that might be labeling the systems under different appellations or in the wider framework of digitalization of maritime operations [29].

2.3. Inclusion and exclusion criteria

Some inclusion and exclusion criteria were used to eliminate irrelevant research papers and focus on the articles relevant to the study. The search criteria for both the Scopus and Web of Science databases were as follows:

- Publication Years: 2013 to 2023
- Document Type: Articles only
- Language: English
- Source Type: Journals
- Publication Stage: Final
- Subject Areas/Research Areas:
  - Scopus: Engineering, Computer Science, Social Sciences, Environmental Sciences
  - Web of Science: Engineering, Transportation, Computer Science, Business Economics, Environmental Sciences

The review period from 2013 to 2023 covers the entire duration that blockchain technology appeared as a remedy for maritime logistics issues starting from the Bitcoin white paper release in 2008 through the emergence of industrial blockchain applications in subsequent years [6]. The review only included peer-reviewed journal articles for strict adherence to scientific methods while omitting conference papers, book chapters and other types of documentation. We focused our assessment on mature research after implementing a peer review process for proper assessment. The chosen subject topics demonstrate the interdisciplinary aspects of blockchain application in maritime systems by covering Engineering and Computer Science for technical implementations and Business Economics and Social Sciences and Environmental Sciences for broader effects.

2.4. Data extraction and synthesis

The search for articles in Scopus yielded 402 articles, while in Web of Science, 191 articles were obtained. To increase the readability and justifiability of results, future research should produce comprehensive queries that were used in every database and guidelines that were used in each stage of screening. Of all the papers identified through the Scopus and Web of Science databases, 113 and 95 articles were included, respectively. The search outcomes from both databases were then summed up to give a total of 208 articles. The process of removal of duplicates and the articles used in the analysis totaled 156. The initial search returned 156 articles, which were screened based on their relevance to the study's objectives and focus on MSW and PCS in the port area. The first step involved the elimination of papers that were not related to the current study. The selected articles for the bibliometric analysis and systematic literature review were 51 articles published from 2013 to 2023. We gathered detailed data from 51 chosen articles which included bibliographic information about authors and publication year along with journal information as well as number of citations while also recording the methodological approach and theoretical frameworks employed and blockchain application contexts for MSW/PCS together with blockchain technology specifications and benefits and challenges identification and key research outcomes. An elaborate extraction method let us conduct both statistical bibliometric evaluations along with analytic thematic study. The researchers used a predefined coding system to code the articles independently in order to improve reliability while a third researcher settled any disagreements. The method minimized researcher bias through a rigorous approach which successfully covered the entire literature domain [30].

For a proper structure of the data extraction, we designated a protocol consisting of 14 specific categories of information for each article: (1) study identifier, (2) full title, (3) authors, (4) publication year, (5) journal/source, (6) number of citations, (7) main methodology (qualitative, quantitative, conceptual, or mixed), (8) theoretical framework, (9) blockchain application context (MSW, PCS, or both), (10) type of blockchain technology described, (11) identified advantages, (12) challenges or limitations, (13) key results, and (14) recommendations for future research. This approach was structured in a way to meet the goal of the extraction of the data in order to focus on different sections while maintaining the cohesiveness of the data. The process of executing cluster classification coding was also done in three stages. First, two authors empirically reviewed each article separately using a set of themes and subthemes derived from the five groups mentioned earlier; secure data sharing, strategic dynamics, success factors and issues, adoption in container shipping sector, and IoT impact on logistics. Each dimension was scored on the three point ordinal scale (0 = not applicable, 1 = partially applicable, 2 = strongly applicable) in order to get the membership of the cluster.

2.5. Bibliometric analysis tools

To perform and display the outcomes of the bibliometric analysis, we have used VOSviewer version 1.6.20, developed by Van Eck and Waltman (2022). This tool allowed visualizing the co-authorship networks, keyword co-occurrence networks, and citation networks, which would help in understanding the current trends in the research, collaborations, and themes under the blockchain domain applied to MSW and PCS. Fig. 1 presents the research protocol of this study, which outlines the general methodological plan of the research. The software VOSviewer earned selection because of its outstanding capabilities in bibliometric network visualization and its broad application in bibliometric research [31]. The software system let us create and examine networks for co-authorship alongside co-citation as well as bibliographic coupling and co-occurrence of keywords and citation relations. The analysis required authors to possess at least two documents for co-authorship evaluation to select leading researchers within the field. A two-occurrence threshold served in keyword co-occurrence analysis to detect major thematic clusters as well as filter out unessential terms. Multiple exploratory analyses led to selecting these parameters which created both an extensive and easily interpretable outcome [32]. The clustering algorithm of VOSviewer uses combined mapping and clustering techniques to recognize separate research communities and thematic subjects in the scholarly literature [33].

3. Bibliometric analysis

This paper aims to identify the research patterns on the use of blockchain technology in Maritime Single Window and Port Community Systems. This study synthesizes 51 articles from peer-reviewed journals by 170 authors from 107 universities during 2013 and 2023. It is evident that collaboration is popular with 94.1 % of the articles being co-authored and only 5.9 % being single-authored. As depicted in Fig. 2, the trend of the published studies in this field reveals that the first work was conducted in 2018. Although the number of outputs provided was still low, the graph identifies that the number of published articles has been rising since 2019, indicating a growing interest in this research area. Recent blockchain-related research in maritime operations began in 2018 because of the wider shipping and logistics industry move towards digital transformation [29]. Academic interest in the subject increased after the 2018 launch of TradeLens by Maersk and IBM due to industry initiatives [34]. The rising number of publications from 2019 onward shows how blockchain recognition grows stronger because researchers understand its ability to solve maritime industry challenges in documentation and trust and information exchange [19]. The majority of articles (94.1 %) exist as collaborative works that show how

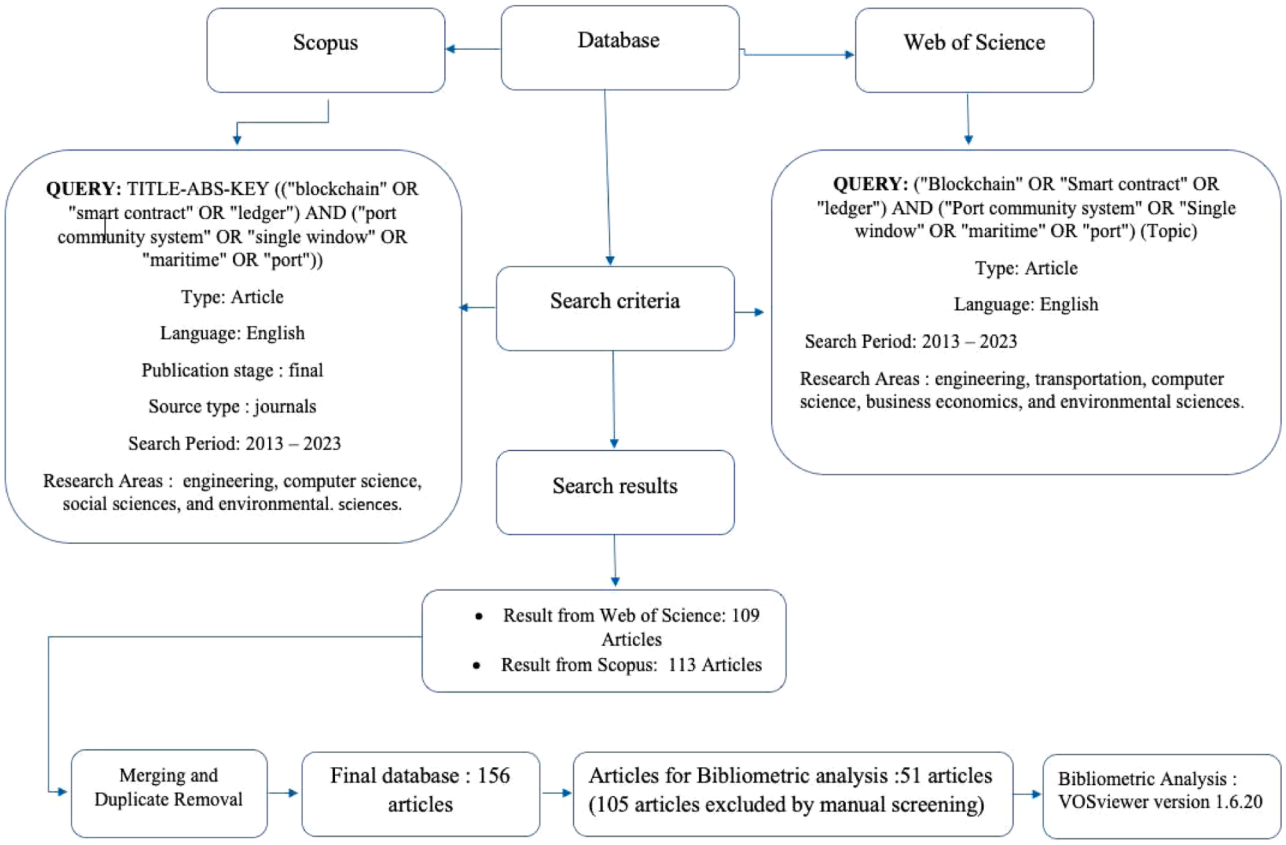


Fig. 1. Bibliometric analysis framework.

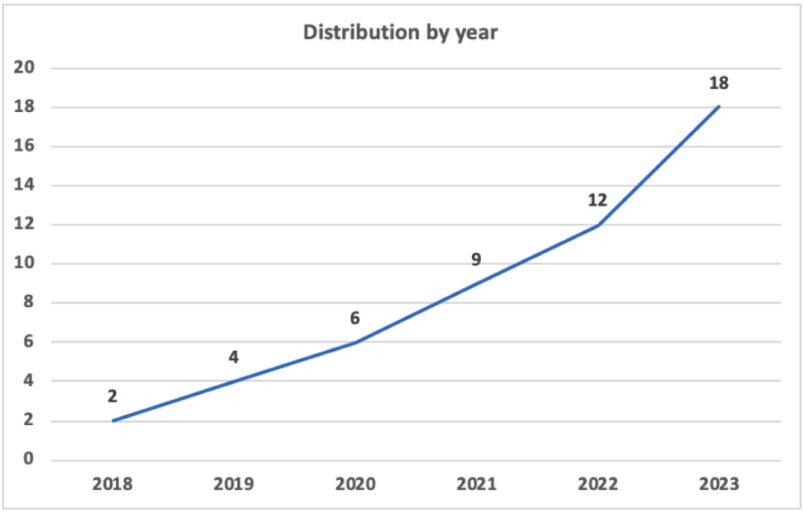


Fig. 2. Publications per year.

researchers with expertise in maritime operations together with computer scientists and business process managers work on this field due to its interdisciplinary nature [9].

3.1. Publications per authors

Analyzing the 51 articles with the help of VOSviewer, it is observed that at least 14 authors have contributed more than two articles concerning implementing blockchain technology in the maritime single window and port community systems. The information on the authors, including the number of their papers, the citation number, and the total link strength, is given in Table 1. The first author Liu Jiaguo has three papers and 136 citations, showing high research output. Jayaraman Raja and Salah Khaled submitted three papers each, receiving 110 citations, proving the authors' high impact. Three authors submitted two documents, namely Loh Hui Shan, Soh Ying Shan, and Yuen Kum Fai, who were cited 161 times, indicating a high academic impact. The total link strength is also different, with Loh and Soh at the highest, with six links indicating a strong collaboration network. This data focuses on identifying the authors and their productivity in the academic world concerning the research field. The authorship analysis demonstrates essential findings regarding scholars who focus on blockchain

Table 1
Authors' publications details.

Author	Documents	Citations	Total link strength
liu, jiaguo	3	136	0
jayaraman, raja	3	110	3
salah, khaled	3	110	3
bauk, sanja	3	21	2
loh, hui shan	2	161	6
soh, ying shan	2	161	6
yuen, kum fai	2	161	6
zhou, yusheng	2	161	6
abdallah, rim	2	118	4
bertelle, cyrille	2	118	4
duvallet, claude	2	118	4
nguyen, son	2	117	0
irannezhad, elnaz	2	38	0
kapidani, nexhat	2	20	2

applications in maritime systems. Studies show that highly cited research occurs predominantly among authors at Dalian Maritime University and Singapore's research centers (particularly those from Asian institutions) where centers of excellence focus upon this domain [21]. Author collaboration such as Loh Hui Shan and Soh Ying Shan supports accelerated knowledge growth in emerging fields because of their high total link strength (TLS=6) [35]. Mutual research projects between academic and industry collaborators generate transformative findings in maritime operations and supply chain management which would be difficult to achieve through independent work [20].

The publication patterns from the most productive authors demonstrate significant specialization characteristics in their writing. The authors achieving the most citations choose to specialize their blockchain research in maritime application areas. Zhou, Yuen and colleagues mainly examine factors for adoption and strategic implementation aspects while Liu's research concentrates on data exchange protocols and technical architecture creation [21,36]. The specialized nature of research reflects the logical split of blockchain science into separate sections including operational technology development and business framework creation and adoption strategy development [8]. The influential research group consisting of Loh, Soh, Yuen, and Zhou developed an evaluation framework which successfully brings blockchain implementation ideas from theory to real-world maritime needs [19].

3.2. Most relevant institutions and countries

From the analyzed 107 institutions involved in the research of blockchain technology applications in maritime single window and port community systems, five organizations have published more than one paper, which equals 12 out of 51 reviewed papers. Four papers (7.3 %) were published by the School of Maritime Economics and Management at Dalian Maritime University, one of the best universities in China focusing on maritime education and research, which has received 154 citations, thus underlining its contribution to the study. Other top institutions are the ASTAR Institute of High-Performance Computing (IHPC) and the Agency for Science, Technology and Research (ASTAR); both from Singapore submitted two articles with 117 citations. Khalifa University of Abu Dhabi United Arab Emirates also has the Department of Electrical Engineering and Computer Science, and the Department of Industrial and Systems Engineering, which has also published two articles, each with 17 citations. The institutional analysis demonstrates that East Asian institutions especially those in China and Singapore have established themselves as leaders in research related to maritime blockchain technology. These economies show such high levels of concentration because maritime trade functions as their top strategic priority while they heavily invest in digital port solutions [37]. The extensive academic reputation of Dalian Maritime University in maritime blockchain research is reflected by its contribution of 7.3 % (154 citations) through its scholarly work addressing technical implementations and business model developments [38]. The high research output from Singaporean institutions demonstrates the influence of national digital strategies because the nation supports Smart Port initiatives and government-backed blockchain projects [39]. Research in blockchain maritime requires specialized departments including engineering at Khalifa University because it combines expertise from computer science, logistics and maritime operations fields [40].

Table 2 highlights the top ten countries that have published works on the application of blockchain technology in maritime single window and port community systems, based on an analysis of 51 documents from Scopus and Web of Science databases. China stands out as the most productive nation, having contributed ten studies with a total citation count of 281, thus showcasing the country's research productivity. The next country with the most documents is Singapore, which has five documents and a high number of citations, with 273, which points to the high impact factor of the country's research output in the field. Although having only three papers, Taiwan scores 262 citations, which allows for the conclusion that its research is highly cited and important. Furthermore, the UK and UAE have just three documents each, but the UK documents have 135 citations, while the UAE has only 110, which shows the relatively higher research impact and visibility of UK documents. The geographic data showcases maritime commerce significance on a worldwide scale and the regional commitment to digital transformation. The significant number of papers with 281 citations from China corresponds to its Belt and Road Initiative efforts and substantial port and digital technology investments [15]. Research conducted in Singapore and Taiwan has disproportionately affected the field with their high citation numbers even though each country published fewer articles likely because they directed their studies toward practical usage of blockchain in maritime settings [41]. Many Middle Eastern nations particularly UAE and Saudi Arabia demonstrate increased blockchain adoption as part of their national port transformation plans [42]. Traditional maritime nation states such as Germany and the USA produced limited research contributions (2 papers each) which demonstrates that dominance in conventional maritime operations does not directly correspond to leadership positions in digital innovation studies for this sector [8]. Transnational impactful research stems from nations which maintain superior academic-industry linkages as well as implemented digital transformation agendas [43].

3.3. Top journals

This review aggregated 51 papers from 37 scientific journals, with 12 journals containing multiple articles on blockchain technology in maritime single window and port community systems. Maritime Policy & Management was identified as the leading journal with the publication of four papers, 178 citations and 10 total link strength, which indicates the journal's high academic productivity and collaborative efforts within the discipline. Research publications spread across different journals help establish blockchain maritime research relationships within academic publications. Data shows Maritime Policy & Management publishes four blockchain papers which have accumulated

Table 2
Most relevant countries.

Country	Documents	Citations	Total link strength
China	10	281	8
Singapore	5	273	4
Italy	4	93	3
Taiwan	3	262	2
UK	3	135	3
UAE	3	110	0
Saudi Arabia	3	48	2
USA	2	88	2
Germany	2	65	1
Estonia	2	64	2

178 citations while demonstrating large policy and management interest instead of technical matters [44]. The research direction emphasizes that maritime blockchain implementation faces organizational along with regulatory barriers instead of technological limitations [21]. Research on blockchain in maritime contexts appears in various publishing outlets that specialize in maritime fields as well as technology and business studies and sustainability topics [9]. Transportation-focused journals are now citing blockchain at high rates because experts view it as a chain-wide transformative technology beyond its role as just a port innovation [45]. The Journals of Marine Policy and Research in Transportation Business and Management also played a critical role, as they published three documents each. Marine Policy garnered 215 citations and a link strength of 7, while Research in Transportation Business and Management accumulated 288 citations and a link strength of 3. Transportation Research Part E: Logistics and Transportation Review stood out with two highly-cited documents (288 citations) and the highest link strength of 13, meaning that the articles in this journal were referenced by many researchers from different institutions. Other journals like Smart Cities and Sustainability from Switzerland, Marine Science and Engineering, Pomorstvo, Ocean and Coastal Management, IEEE Access and Journal of Maritime Research have also been relevant. These journals have different abilities of impact and interconnection based on the citation frequency and link strengths. The main emphasis of these publications is on computer science, engineering, and transportation. The spread of journal publications demonstrates blockchain maritime research improving from its beginning conceptual stage towards its adoption as specialized applied applications. The initial research period (between 2018 and 2019) found its publications in general transportation and management academic journals that proposed conceptual ideas and hypothetical usages [19]. Recent journal publications demonstrate better implementation depth through their publication in specialized journals like IEEE Access which supports detailed technical details [46]. The appearance of blockchain maritime research in Sustainability and related sustainability-focused journals demonstrates that scientists now dedicate attention to blockchain's sustainability benefits including reduction of paper processes and optimized logistical operation [9]. Technical and domain experts from marine and computer science engineering fields are working more closely together to tackle implementation problems in their respective journals [47]. The journals above show that the research on blockchain applications in MSW and PCS is multidisciplinary. Thus, the research implies increasing interest in this field, focusing on publications in journals related to maritime policy, transportation, and logistics.

3.4. Most impactful authors

When focusing on the set of the most productive authors in the given dataset, it is possible to identify several authors with the highest citation, documentary, and collaborative linkages. Soh Ying Shan, Yuen Kum Fai, and Zhou Yusheng have the highest citation count of 161 papers with 2 papers each, further stressing their high impact on the research. Another scholar, Liu Jiaguo, has 3 documents accounting for 136 citations. Jayaram Raja and Salah Khaled have the highest number of publications, 3 documents each and 110 citations, making them one of the most noticeable scholars. The academic influence based on citations demonstrates superior quality over extensive output within this evolving branch of study. The targeted research performed by Zhou, Soh, and Yuen has delivered 80.5 citations for each of their papers which exceeds the 36.7 citations accumulated by Jayaraman and Salah in their publications [21]. The research approach of authors differs according to their background since technical professionals (Liu and Salah) develop implementation frameworks but maritime business experts (Yuen and Zhou) focus on adoption models and business value propositions [46]. The combined knowledge between technical and maritime professionals has produced an extensive evaluation about blockchain applications within maritime operations through their combined examination of system requirements and organizational benefits [48]. The industry requires actual implementation outlines and evaluation structures which render research articles with these elements more impactful based on citation metrics [21]. Also, the collaboration and networking power is enhanced by the fact that Soh, Yuen and Zhou have a total link strength of 6, thus implying that they have strong research collaborations. Similarly, Bertelle Cyrille and Duvallet Claude have a link strength of 4, and Jayaram and Salah have a strength of 3, showing that connected works can help increase research output. The Network analysis demonstrates structured research partnerships between scholars with suitable research specialties. Through their collaboration the research team Soh, Yuen, Zhou achieves the highest TLS rating of 6 because they combine their expertise in maritime economics with technology adoption and strategic management to solve blockchain implementation problems simultaneously [38]. Researchers using various disciplines within their work can conduct broader examinations than what stands alone through individual fields of expertise. The European collaborative cluster (Bertelle, Duvallet, Abdallah with TLS=4) merges specialists in computer science, system architecture and maritime logistics among others [49]. The most important research papers demonstrate an international collaboration pattern because multiple authors from different countries work together thus forming stronger outcomes according to research [21]. The networks built for collaboration function as mechanisms that facilitate knowledge sharing between researchers who originate from advanced technical regions and major maritime centers to develop solutions with practical applications [50]. Therefore, identifying the most productive authors in the present dataset demonstrates the significant role of several eminent scholars who have significantly influenced the development of the field through their numerous publications, extensive citations, and strong collaboration with other researchers. These authors are a clear example of how collaboration between scholars is crucial and how crucial it is to strengthen research collaborations to increase the visibility of the research. Thus, it can be concluded that the works of these influential authors will remain relevant and inspiring for the further development of the field.

3.5. Author keyword co-occurrence

The author's keyword co-occurrence analysis was conducted to map the intellectual domains related to blockchain application in maritime single windows and port community systems. Using VOSviewer with a minimum threshold of two occurrences, 18 keywords were extracted and illustrated in Fig. 3, revealing five distinct clusters. The keyword co-occurrence analysis generates a semantic framework to illustrate how different literary concepts relate with each other in academic research. The algorithm in VOSviewer recognizes five distinct clusters demonstrating blockchain research in maritime settings possesses multiple conceptual bases and application fields in maritime [31]. The cluster method enables researchers to spot dominant themes while pinpointing vacant areas in available literature research. The literature presents technical and operational elements well while regulatory and standardization matters surface as new problems that lack direct relations with primary blockchain applications [10]. The position of keywords in the network model displays conceptual relatedness because literature content shows co-occurrences between nearby keywords that help generate new research ideas [32]. The term 'blockchain' is the most significant in this line of scholarly inquiry, reflecting its fundamental importance, appearing 37 times and referred to 66 times. The first cluster, "information sharing" (2 occurrences, 5 links), "innovation" (2 occurrences, 4 links), "ports" (2 occurrences, 3 links), and "smart contracts" (5 occurrences, 11 links), gives an insight into core aspects of integration of blockchain in ports. Cluster 2 presents more complex concepts and specific reference to industries, comprising "decentralized technologies" (2 occurrences, 8 links), "game theory" (2 occurrences, 3 links), "smart port" (6 occurrences, 15 links), and "shipping industry" (7 occurrences, 18 links). The third cluster focuses on turning to industry

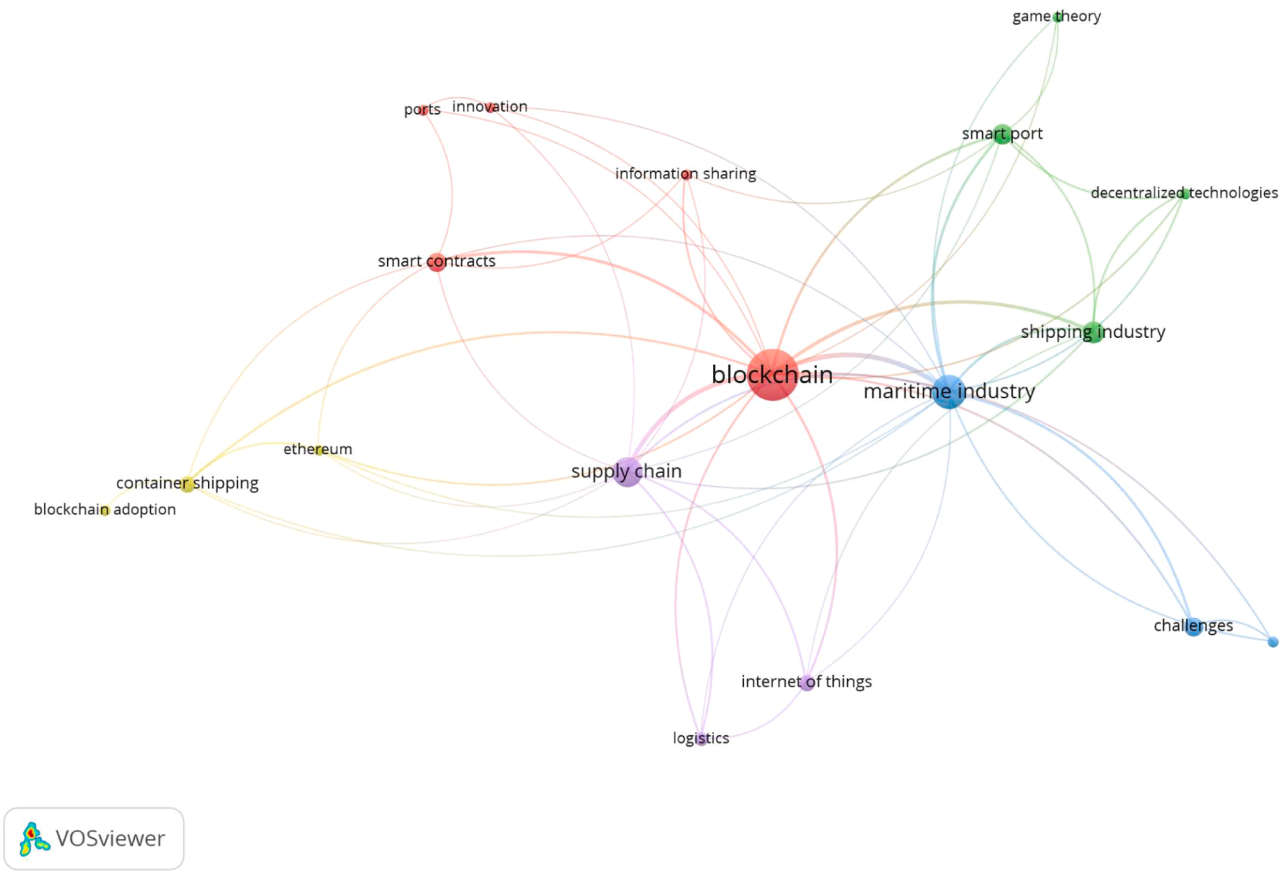


Fig. 3. Keywords matching (elaborated with VOSviewer).

challenges and use such specific terms as "challenges" (5 occurrences, 9 links), "critical success factors" (2 occurrences, 6 links), and "maritime industry" (17 occurrences, 40 links). As for Cluster 4, it is dedicated to using blockchain in container shipping with such tags as "blockchain adoption" (2 occurrences, 1 link), "container shipping" (4 occurrences, 9 links), and "ethereum" (2 occurrences, 7 links). The fifth cluster of topics of the journal relates to logistics and supply chain, featuring "internet of things" (4 occurrences, 11 links), "logistics" (3 occurrences, 9 links), and "supply chain" (13 occurrences, 29 links) as key elements. These keyword clusters reveal important patterns in the development of the concept of the blockchain with reference to the sea-related field and capture both the initial and the further level of thoughts about it. The occurrences of "blockchain" at 37 points with 66 links validates its position as the fundamental technological element that integrates various areas of application domain [36]. Studies demonstrate that researchers deploy blockchain as maritime industry-specific solutions instead of generic technology platforms through their strong frequency of block-chain-maritime industry connections (17 occurrences and 40 links) [45]. The independent concept "smart contracts" with five appearances and eleven connections demonstrates the necessity of automation features that surpass basic distributed ledger operations [46]. The numerous occurrences of "supply chain" (13 times in total and 29 links) prove blockchain is mainly examined as an instrument for supply chain management in maritime domains because it allows full visibility and traceability between all stages of delivery [10]. Research on applications for container shipping through "ethereum" has advanced to the evaluation stage as shown by cluster 4 containing "ethereum" twice and seven external connections [36].

4. Research clusters

The five clusters identified in the current section were derived from a mixed approach consisting of bibliometric and comprehensive thematic analysis. The process of clustering proceeded in three stages: (1) The initial identification of the clusters with the help of the VOSviewer using keywords; (2) The manual identification of the concrete thematic contents through an analysis of articles; (3) The validation process that took place with the three main researchers of the study. The works were assessed concerning the specific five themes that are thematic clusters; it was possible for an article to be grouped under multiple themes in case it addressed different aspects of the selected themes. This integration of the two study strategies further paved the way for devising an all-embracing taxonomical map of the field which embraced the statistical relationships revealed by the bibliometrics study and the conceptual embedding postulated from the internal qualitative analysis. Based on the literature review, the following five major research clusters were derived concerning blockchain technology implementation in the context of maritime single window and port community systems. The first cluster is associated with the opportunities for secure data sharing and innovations based on the blockchain technology in the environment of the ports and the second cluster addresses the strategic and operational effects of the integration of the decentralized technologies including the blockchain in the concept of smart ports. The third cluster deals with the main barriers and drivers to using blockchain technology in the maritime sector. The fourth cluster further refines the investigation to the application and effects of blockchain technology in the container shipping industry. Last, the fifth cluster is focused on the opportunities of using blockchain as an enabler of the Internet of Things and supply chain activities to revolutionize maritime logistics. The categorization of these five separate research clusters serves as an

important field advancement because it creates a systematic classification system which directs academic research and industry operationalization [10]. The cluster analysis method conforms to main techniques used in bibliometric analysis to detect intellectual boundaries in developing research fields [32]. The natural development of blockchain research in maritime settings shows itself through linked research clusters which start with secure data sharing principles before transitioning to container shipping applications and finishing with IoT technology integration. The implementation challenges of blockchain consist of both technical elements found in clusters 1, 2 and 5 and organizational/strategic aspects located in clusters 3 and 4 while demonstrating the socio-technical characteristics of blockchain [29]. A comprehensive viewpoint constitutes the essential foundation for examining how technological competencies unite with industrial implementation factors during the blockchain industry evolution in maritime operations [8]. (Tables 3–5)

This distribution analysis clearly shows the positioning of research themes: The largest share of the work is taken by the challenges and success factors (C3) comprising 41.2 % of the work and contributing the most to the visibility and citations. However, there is a marked increase in publication activity across all the clusters from the year 2020, which supports the increased focus on the application of blockchain technology in the maritime industry over the past few years.

4.1. Blockchain-enabled secure data sharing and innovation in port

The use of blockchain technology can help secure data sharing and bring innovation in the port environment. An example is to provide a common and secure ledger that can be accessed by multiple parties in real-time and minimize the need for third parties [22]. This can help impose control on export customs for containers through pre-notification systems and transferring the export declaration information in electronic format [22]. Due to the decentralized nature of blockchain, users can perform online transactions at a lower cost than conventional intermediaries and with the use of anonymous alphanumeric addresses instead of sensitive personal details [22]. Innovative solutions specific to the maritime industry involve using automated systems in monitoring the ports, handling cargo data, tracking containers and managing ships' stowage [22]. The blockchain application in port supply chains can be based on the openness and non-alterability of the blockchain records [24]. In ports, blockchain technology can enhance data protection and confidentiality in serving ships by using characteristics such as decentralization, non-tampering, openness, and transparency [25]. Some of the features of blockchain such as smart contracts enhance effective logistics process management and supply chain supervision [24]. Implementing the Blockchain-based port community systems brings about the following strategic advantages: better synchronization, less information gap, and lower transaction costs [26]. This is because through tokenization of payments, cross border transfers can be done easily across the trade parties without the need to worry about exchange rates [26]. The technology addresses the problem of trust among the port players as it operates in a peer-to-peer network rather than depending on third party commercial

Table 3
General distribution of studies by cluster.

Cluster	Cluster Title	Total Studies	Studies Exclusive to Cluster	Shared Studies	Average Citations
C1	Blockchain-Enabled Secure Data Sharing and Innovation in Ports	17	7	10	42.8
C2	Strategic Dynamics of Decentralized Technologies in Smart Ports	14	3	11	36.2
C3	Challenges and Success Factors for Blockchain in the Maritime Industry	21	8	13	68.4
C4	Blockchain Adoption in Container Shipping	15	5	10	44.3
C5	Transforming Maritime Logistics through Blockchain, IoT, and Supply Chains	10	2	8	28.9
Total		**77 (total entries)**	**25 (single cluster)**	**26 (multi-cluster)**	

Explanatory note: Out of 51 studies, 25 are classified in a single cluster, while 26 belong to multiple clusters simultaneously, explaining why the total number of entries across clusters (77) exceeds the number of unique studies (51).

Table 4
Temporal evolution of publications by cluster.

Year	C1: Secure Data Sharing	C2: Strategic Dynamics	C3: Challenges & Success Factors	C4: Container Shipping	C5: IoT & Supply Chains	Total
2018	0	0	1	1	0	2
2019	0	1	2	1	1	5
2020	5	4	6	3	2	12
2021	5	3	6	3	3	14
2022	5	4	4	4	3	12
2023	2	2	2	3	1	6
Total	17	14	21	15	10	51*

* The sum of studies by cluster (77) exceeds the total number of studies (51) because some studies cover multiple clusters.

Table 5
Dominant thematic focus in each cluster.

Cluster	Dominant Sub-themes	Representative Articles [citations]
C1: Secure Data Sharing	• Data security architecture • Validation mechanisms • Port information sharing • Distributed trust	[22] Zhou et al. (2020) [215] [53] Irannezhad (2020) [38] [50] Li & Zhou (2021) [45] [44] Ahmad et al. (2021) [34]
C2: Strategic Dynamics	• Port digital transformations • Supply chain integration • Decentralized models • Smart Ports	[38] Liu et al. (2022) [37] [45] Philipp et al. (2019) [42] [4] Gao et al. (2021) [27] [54] Kuo et al. (2022) [28]
C3: Challenges & Success Factors	• Adoption barriers • Critical success factors • Regulation and compliance • User acceptance	[20] Yang (2019) [288] [22] Zhou et al. (2020) [215] [15] Pedersen et al. (2019) [112] [52] Balci & Surucu-Balci (2021) [54]
C4: Container Shipping	• Container traceability • Bill of lading digitization • International trade • LCL operations	[20] Yang (2019) [288] [21] Irannezhad & Faroqi (2023) [38] [48] Elmay et al. (2022) [27] [43] Tan et al. (2018) [24]
C5: IoT & Supply Chains	• IoT-blockchain integration • Process automation • Real-time data • Cold chain monitoring	[67] Wang et al. (2021) [59] [68] Yang et al. (2021) [56] [63] Le et al. (2023) [7] [48] Elmay et al. (2022) [21]

entities [22]. By adopting blockchain in ports, the maritime processes could be enhanced as well as the physical movement of the goods [22]. It allows sharing of data that cannot be forged in real time, quicker and efficient clearance at the customs and/or other procedures [22]. Certain applications such as Trade Lens, a product of a joint venture between IBM and Maersk, ensure that logistic transportation and shipping is fully trackable using blockchain technology. These innovations show that blockchain has the possibility of revolutionizing data sharing and improving operations in port areas. Challenges that hinder the use of blockchain in ports include low digital literacy among the actors, unavailability of consensus standards, and the reluctance of operators to engage and share information [22]. However, the opportunities that come with the secure exchange of data and innovation resulting from the use of blockchain technology are much more significant than the challenges mentioned above. With the current trend of port transformation and digitalization, blockchain will be the key driver for the improvement of the efficiency of the port through the competitiveness of the industry. The secure data sharing elements of blockchain technology transform port information systems beyond their traditional problems with data fragmentation and trust issues according to [34]. The recognized advantages conform to information systems theory about how reducing information asymmetry works in diverse stakeholder systems [19]. This cluster shows an academic development where researchers moved from abstract conceptual models toward dedicated operational designs of dispersed privacy-protected blockchains [36]. Research demonstrates how blockchain can improve data reliability and expense reduction but current evidence does not show its real-world effectiveness at ports [51]. The majority of research in this cluster demonstrates a technological determinism approach because it assumes blockchain technical capabilities will produce business value without proper consideration of implementation-level socio-technical complexities [15]. TradeLens serves as an example of adaptive strategy practiced by the industry for the selection of a few dominant platforms, where the execution of the platform started in 2018 and stopped in 2023. This will help highlight the future research agenda that is to identify how power dynamics would play out in these ecosystems and governance structures [50]. Studies that investigate blockchain implementations' GDPR (General Data Protection Regulation) compliance provide limited assessment of data protection frameworks even though this topic is often mentioned in research [42].

4.2. Strategic dynamics of decentralized technologies in smart ports

Smart ports can be defined as the adoption of technologies and big data analysis in the management of ports. The main elements of smart ports are automation, the Internet of Things (IoT), artificial intelligence and blockchain [16]. Blockchain technology is quite useful in supporting a secure data-sharing environment in the port ecosystem since it is a decentralized system [27]. It can facilitate trust in information sharing by using an expandable protocol to store information, allowing only permitted users to access it [28]. It resolves the bottleneck of traditional centralized storage, effectively solves problems of mutual trust between enterprises, facilitates the supervision of cold chains by relevant law enforcement departments, and enables complete real-time traceability of port cold chains [28]. Blockchain and smart contract technology also provide small and medium-sized enterprises (SMEs) opportunities to integrate cross-organization business processes by creating open and coordinated information environments [28]. Blockchain increases smart port potential by creating a reliable and irreversible system of record-keeping that helps to protect data from fraudulent manipulation [29]. It establishes communication between the port authorities, shipping companies, freight forwarders and customs, minimizing the manual work and paperwork [30]. Using smart contracts based on blockchain technology helps automate the logistics processes and self-executing contracts, thus improving efficiency and eliminating delays [31]. Some benefits of adopting blockchain in smart ports are that it increases supply chain transparency, improves collaboration between all parties involved, and increases port operations' productivity [32]. Hence, through the implementation of blockchain, there is no replication of data and the tracking of cargo movements is done in real-time, thus minimizing the chances of fraud and occurrence of errors [33]. It also supports integration with IoT devices through which data can be automatically captured and smart contracts can be executed based on specified conditions [34]. Blockchain enhances container activities coordination, increases customs authorities' effectiveness, and optimizes intra-terminal transport flows [22]. It also enables some automation in the services that include arrival notice, cargo pick up and delivery booking, thus reducing the time taken and enhancing the planning of the port operations. Maritime digital transformation literature received significant progress through the identification of blockchain as a smart port development's essential strategic element [38]. The cluster demonstrates blockchain takes on a new position within companies since it represents part of an established technology framework combining IoT devices with artificial intelligence and automated platforms [52]. Strategic management research dominates the field of blockchain studies because scholars adopt digital business strategy and platform ecosystem frameworks to evaluate blockchain transformation potential [53]. Research conducted under these parameters shows a broader approach than other clusters by recognizing that various technologies unite to construct complete smart port solutions [54]. This research has an important drawback because it lacks proper assessment methods to measure the actual strategic value blockchain implementations produce [19]. Studies mainly use conceptual models instead of objective performance evaluations or ROI assessments to show theoretical advantages [43]. The existing literature knows about governance complications that emerge in managing intricate technology networks but lacks systematic research on defining authority systems for blockchain implementations across various stakeholders in port operations [10]. The adoption of permissioned blockchain architectures within smart ports happens because of practical motivations to enhance scalability alongside confidentiality while generating concerns about operational control versus decentralized principles which need better examination [28]. Similarly, the same potential benefits of blockchain-based systems can be observed in yard straddle carrier and tractor operations through registration and recording of the data including the location, power, weight, fuel consumption and performance [30]. It can be noted that the absence of a mediatory structure in the blockchain system minimizes transaction costs and accelerates the process of transactions [22]. It allows for direct communication between port stakeholders, fostering collaboration and innovation [35]. Smart ports leveraging blockchain technology can achieve higher levels of automation, improved resource utilization, and enhanced customer service [36]. However, some issues can be associated with the application of blockchain in smart ports which include the challenges of system integration, the scalability challenge and the legal frameworks that govern the use of the technology [37]. To address these challenges, cooperation between the interested parties and efforts to create standards and good policies are critical [38]. With the development of smart ports in the future, it is important to understand how to apply the blockchain in strategic planning to enhance the performance of the maritime industry.

4.3. Challenges and success factors for blockchain in the maritime industry

The maritime sector is generally risk-averse, tending not to be an early adopter of new technology [39]. Some stakeholders prefer to keep their data secret, as competition is fierce and numerous players compete with the same service [39]. The large amount of data and traffic generated in blockchain requires wideband channels like 5G or 6G, while the internet speed and stability at sea are usually lower than ashore [39]. Additionally, blockchain technology is high-energy

consuming and causes a high carbon footprint [39]. Thus, both the awareness and the knowledge of blockchain are still low, and there is a need to create educational, training, and capacity-building programmes at the regulatory and administrative levels and among operational staff [39]. . Despite the advantages that blockchain technology presents to the maritime industry, various challenges hinder the adoption of the technology. A major concern is on the scalability problem whereby as the number of transactions rises, it becomes time-consuming and consumes more energy [40]. The compatibility problem is associated with the lack of common standards that would exist globally, which hinders the interaction of different blockchains and the exchange of information between them [41]. Legal and compliance issues are another challenge across jurisdictions; thus, the application of blockchain in the maritime sector is challenging [41]. The mentioned cluster provides one of the most productive views on blockchain implementation through a focus on substantial adoption barriers in maritime operations [21]. Research experts in this field address crucial social and technical limitations which must be solved before blockchain initiatives can succeed [55]. The unique characteristic of this field is using innovation diffusion theory with technology acceptance frameworks to study adoption barriers while supplying theoretical foundations which differentiates it from other research clusters [56]. Classifying maritime culture as risk-averse fits within conventional organizational change theories about innovation resistance in traditional fields [29]. Discussions of some of the technical impediments that blockchain possesses, including scalability and energy consumption provide realism to those rationally motivated indications of developmental chances and improvements beyond what excited literature explains [57]. The distinct features of advanced studies within this cluster are their systematic examination of institutional elements together with governance structures because technical tools alone fail to solve critical trust-based problems and market competition and regulatory matters [49]. The studies in this area still face a major challenge because they focus on barrier descriptions more than developing implementation frameworks that overcome these obstacles as a whole [50]. The field requires more quantitative studies to measure which barriers and testing strategies matter most since current research lacks quantitative evidence [43]. Nonetheless, the absence of well-defined legal protocols and conventions diminishes the technology's popularity [22]. Issues related to the privacy of data and the adherence to the General Data Protection Regulation (GDPR) are also of concern [42]. Nonetheless, several factors favor the integration of blockchain technology in the ports and shipping industries. Some of the reasons are related to the need for better supply chain visibility, better data exchange, and increased efficiency [43]. The benefits of cost-saving, minimizing paperwork and improving business processes are the main incentives for industry participants to consider blockchain applications [44]. The advancement of technologies such as the digitalisation of the shipping industry and the concept of smart ports enhances the adoption of blockchain technology [45]. To address the challenges stated above, possible solutions include creating blockchain systems capable of expansion like permissioned networks and use of off-chain storage systems [46]. This can be achieved through the collaboration of all the players in the industry such as the shipping companies, port authorities, and technology suppliers [47]. Interacting with the regulators and policymakers to set down the legal standards and procedures is important to gain and obtain the trust and acceptance of the participants [48]. For the successful adoption of blockchain in the maritime industry, there is a need to integrate technology, policies, and all the stakeholders involved in the industry [49]. Real-world projects that explain how blockchain can produce value will go a long way in establishing trust and fostering the technology's adoption [50]. More so, educational and training measures should be pursued to close the gap and prepare the marine workforce to harness blockchain technology [51]. The following are other success factors; application of blockchain in conjunction with other systems and technologies such as IoT and artificial intelligence to come up with a comprehensive approach [52]. Therefore, collaboration with experienced technology vendors and their know-how can enable faster development and deployment of blockchain solutions [53]. Governance structures and clear ownership of data properties are important and inevitable prerequisites which must be set to improve the integrity and security of blockchain networks [54]. When a particular industry addresses the challenges and harnesses the opportunities of blockchain technology, there is a need to build cooperation and take a positive approach for the prosperity of the technology. Thus, considering the technical, regulatory and organizational barriers will allow the sector to benefit from blockchain application and support innovation, efficiency and competitiveness in the maritime environment.

4.4. Blockchain adoption in container shipping

Container shipping, a vital part of the global trade, will most likely be improved by the blockchain application. Blockchain provides the possibility to change the current container tracking and management system, increase supply chain transparency, and simplify paperwork [55]. Through blockchain, the participants can obtain information on the movement of the containers in real-time thus increasing efficiency and minimizing the incidence of fraud [56]. An example of blockchain application in container shipping is the digitization of the bill of lading (B/L), one of the most important documents in the supply chain since it is a receipt, a contract, and a negotiable document [57]. Standard paper-based B/Ls are often delayed, incorrect or fraudulent, which causes inefficiency in the shipping process [58]. Electronic Bills of Lading using eBLs based on Blockchain technology provide a secure and efficient solution to transferring ownership of the documents, which can be done in real-time [59]. Blockchain application in container shipping profoundly affects supply chain visibility and transparency. The cluster shows a notable transition from abstract models to concrete maritime applications where container shipping serves as a primary field for blockchain adoption in maritime sectors [59]. This cluster's research studies specific document flows and business processes by examining electronic Bills of Lading because they are severe challenges within conventional container shipping [20]. Research on blockchain for shipping focuses on containers because they represent both a major economic sector and a technology-friendly domain due to procedural and documentation protocol standards [60]. The studies within this cluster follow a more technical direction compared to other clusters while focusing on the development of implementation architectures alongside consensus mechanisms as well as interoperability standards [61]. The field produced valuable findings through its creation of blockchain maturity models for specific industries which aid organizations to evaluate their readiness while establishing proper implementation routes [62]. A major restriction exists because most blockchain research focuses on abstract models instead of real deployment analysis and the creation of evidence about container shipping sector value addition from blockchain solutions [56]. A major deficiency exists in the literature because document digitization benefits from blockchain have been clearly established yet there is insufficient investigation regarding integration methods with existing legacy systems and migration planning [59]. Research on B/L digitization using blockchain demonstrates a preference for incremental changes instead of disruptive innovation since it focuses on enhancing existing procedures rather than producing new business concepts [63]. Thus, by implementing a distributed and cryptographic database, blockchain allows one to track the movement of containers from the initial point to the final one [59]. This increased transparency helps the stakeholders track the movement of consignments in real-time, thus eliminating uncertainty risks and allowing for timely decision-making [60]. Smart contracts can also facilitate other activities such as releasing goods and making payments thus making the supply chain more efficient [61]. Blockchain can considerably minimize the paper work and other activities in the container shipping industry. One of the most prominent

examples of the implementation of blockchain technologies is the TradeLens platform launched by Maersk and IBM companies from 2018 to 2023. This platform demonstrated the possibilities to negotiate and execute trade documents through digital and automated means thus eradicating paper work before the discontinuation of the platform [62]. This reduces the time and cost taken in the process and eliminates chances of making mistakes and inconsistencies [63]. Through blockchain, changes made to a particular document cannot be altered as they are, thus increasing the confidence of stakeholders [64]. By implementing blockchain technology in container shipping, the incidence of fraud can be minimized while security can be improved. Blockchain helps to minimize fraud since it offers a clear and easily traceable history of the container movement and ownership [65]. Smart contracts can also implement certain rules and conditions that are already pre-determined and eliminate the chances of manipulation [66]. Blockchain applications in Container Shipping can also include connection with IoT devices and equipment, such as sensors, RFID tags for monitoring and control of the container and identification of any manipulation or interference [67]. There are other advantages of blockchain in container shipping than just functional enhancements. It can help integrate the parties involved in the data sharing process to avoid creating data monopolies [51]. Blockchain-based platforms may help coordinate several participants, such as shippers, carriers, ports, and customs authorities, leading to a more effectively connected system [27]. This integrative approach thus has the potential to enhance customer service, faster clearance procedures and less time wastage [31]. However, some issues hinder blockchain implementation in the container shipping industry, including standardization, interoperability, and legal issues [37]. To address these challenges, stakeholders in the industry must, therefore, work together, establish standard operating procedures and involve the regulators in the process [47]. Thus, as the industry continues to pursue the integration of blockchain solutions, these difficulties need to be solved, and a proper environment for applying technology should be created. Blockchain has great potential in the container shipping industry, and its successful application can revolutionize the industry and bring positive changes. This is because blockchain can make the process of container shipping easier by digitizing them, increasing the transparency of the process and allowing for collaboration.

4.5. Blockchain as an enabler of IoT and supply chain activities

Integrating blockchain based technology with the Internet of Things (IoT) will also revolutionalise the application of maritime logistics particularly with respects to issues to do with security, transparency, and automation of the entire chain. Thus, blockchain plays a vital role in IoT as it helps preserve the decentralized and immutable platform for the IoT devices to share their data and interact [52]. Thus, integrating blockchain with IoT will allow maritime industry stakeholders to have real-time monitoring, improved tracking, and better handling of logistics [35]. Another major area of using blockchain and IoT in port and ship management is tracking and controlling of cargo and assets. Sensors and RFID tags, considered IoT devices, can be put on containers, vehicles, and equipment with the help of which it is possible to track and gather real-time information on location, temperature, and humidity, etc [30]. This data can be safely shared across a block chain network, thereby providing an end to end record of supply chain as stated in [46]. In this way, smart contracts can be activated by IoT data to execute tasks including customs clearance, payments and insurance claims [56]. Use of block chain technology has particularly applied bundle of efficiencies to supply chain by offering particularity in tracking and tracing of IoT. Through IoT devices and blockchain's distributed ledger, the various parties can monitor the movement of the products from the source to the consumer [62]. This level of granularity ensures that one can detect any anomaly in efficiency, deviation or even problems with quality thus allowing for immediate intervention and management of risks [54]. Due to the characteristics of blockchain technology, the tracking data is protected from the possibility of altering or fake information [29]. In the context of logistics and supply chain, smart Contracts are used for the order fulfilment with the help of blockchain technology. These contracts have inherent power of their self-execution and triggering certain actions upon occurrence of stated circumstances as for example, making payments when goods are delivered [31]. Additionally, through smart contracts, the trade finance transactions will be able to be executed without human intervention and increase the flow of liquidity [55]. Effective and timely management of the supply chain in the maritime industry can be made effective by applying smart contracts through automation of routine processes and eliminating middlemen [53]. The use of these two technologies in the maritime industry creates new opportunities for new business models and services. For instance, given occurrence of blockchain, it is possible to tokenize real objects such as containers or ships. This innovation can create opportunities for fractional ownership and investment and thus open up the traditional markets [38]. IoT data from ships and ports can be sold through the data marketplaces provided through the blockchain which can help the stakeholders to gain valuable information for their improvement and decision-making process [33]. Another area, where blockchain can help create new insurance products is in parametric insurance, which is based on real-time IoT data [45]. Nevertheless, to effectively apply blockchain and IoT in maritime logistics, some challenges need to be considered. Interoperability and standardization are important for the purpose of data sharing and compatibility among different systems and devices [34]. It is imperative to have strong cybersecurity by implementing strong measures to prevent the IoT devices and blockchain networks from cyber threats or malicious attacks [32]. For the effective implementation of blockchain and IoT in cross-border trade and logistics, there is a need to come up with new regulatory frameworks [44]. However, there are numerous possibilities of using the integration of blockchain and IoT in the maritime supply chain logistics as discussed in the following sections. As a technology that allows for secure data sharing, increases efficiency and automates processes, blockchain can be considered as a key enabler of IoT supply chain activities [40]. As the world moves towards the digitalization of the maritime industry, the use of blockchain and IoT cannot be overlooked because they will play an important role in enhancing efficiency, reliability and innovation in the international shipping supply chain [16]. Research in this cluster explores the highest level of technological integration between blockchain technology and IoT systems to establish complete digital supply chain frameworks [47]. This research group differentiates itself by studying technological convergence instead of individual blockchain deployments because it understands that collaborative systems generate maximum benefits [46]. The research within this cluster demonstrates how blockchain operates as a security framework for IoT-based operational data systems that need authentication and protection mechanisms for secure information sharing [64]. Research provides two essential findings regarding architecture design by demonstrating how to solve implementation obstacles that include data validation techniques and consensus mechanism selection and off-chain storage implementation [47]. There are several business models created by the blockchain IoT in the academic literature including data marketplaces, the tokenization of assets and parametric insurance. The following models have the prospect for creating value of a sort that may go beyond the goal of improving efficiency [65]. A major weakness of this cluster becomes apparent through the little focus on security vulnerabilities which emerge when blockchain meets IoT systems because only a few studies demonstrate complete security threat models and architectural designs [42]. The implementation of complex technology systems requires focused attention on organizational readiness elements together with change management strategies because technical integration difficulties already receive sufficient documentation [52]. Research studies often fail to accurately predict the amount of resources and timetable needed to implement blockchain-IoT integration which leads stakeholders to

hold untrue expectations about adoption ease [50]. Experts in this frontier research area are building the foundation for blockchain to operate as the trust base within extensive digital networks that cover the marine supply chain [66].

5. Conclusion

This systematic review of research articles and the bibliometric analysis that supported it for papers pertaining to blockchain implementation in maritime single windows and port community systems provides hard evidence for the significant possibilities and growing literature regarding this application. The examination delineates five principal research clusters: Blockchain-Enabled Secure Data Sharing and Innovation in Ports; Strategic Dynamics of Decentralized Technologies in Smart Ports; Challenges and Success Factors for Blockchain in the Maritime Industry; Blockchain Adoption in Container Shipping; and Blockchain as an Enabler of IoT and Supply Chain Activities. The results indicate that the number of publications has been growing every year since 2018, and the authors, including Irannezhad, E., Shen, L., Liu, Y., and Zhou, Y., have contributed to the reputable publications such as Maritime Policy & Management, Research in Transportation Business & Management, and IEEE Access. With regards to the activity level, China can be regarded as the most active country, and Shanghai Maritime University appears to be one of the leaders of the activity among the major players. Future works should also consider the challenges and barriers of adopting blockchain in the port context as well as provide practical cases that depict the concepts like transparency, collaboration, and trust. This systematic review delivers multiple vital added values to the developing blockchain applications used in maritime single windows and port community systems. The research measurement helps establish five specific research clusters which create a detailed classification system for additional study research [32]. Previous research analyzed maritime blockchain as a single domain but the classification method develops finer distinctions through these clusters [16]. The analysis shows increasing research activity since 2018 which corresponds to Tijan et al.'s [29] findings about quick digital transformation in maritime sectors using blockchain technology as an important enabling tool. Our analysis also supports the key positions of China and Singapore, as both countries identify digitalization at sea as important as the former has reported positive progresses in intelligent port layout and the latter has taken concrete actions toward digital trade promotion [37]. Researcher and institutional prominence signify the development of specialized knowledge networks according to Kumar and Iyengar [35] in emerging technological fields at their early consolidation phase. Due to the increased evolution towards smart ports, it is important to analyze the additional value resulting from the integration of blockchain and the Smart port can be regarded as one of the latest success stories in the field of intelligent management. The findings of this study have implications for future research studies and industry professionals in the following ways. It contributes to the theoretical development, presents a literature review and presents future research directions. For the industry, it provides recommendations that can help determine the best strategies of integrating blockchain in enhancing port operations and smart port development. However, the study also highlights the main barriers to the spread of blockchain technology including technical factors like compatibility and expansion, lack of legal framework, standards, high costs of deployment along with End user resistance. There is however an imperative call to endeavor to address these challenges through the concerted effort of all the stakeholders who include the industry, the governmental organizations and academic institutions. Based on the research study, recommendations are made to the researchers and the industry players. Firstly, there is a need for standard architecture and processes that may help identify, integrate, and implement blockchain in the maritime industry in order to overcome the challenges of compatibility and expansion. Second, case and pilot studies should be carried out to outline real-world implementation problems, and develop effective approaches to resolve them. Thirdly, there is the need to examine the legal and regulatory issues of the use of blockchain technology, particularly in the international maritime trade, to develop appropriate frameworks and guidelines. Therefore, understanding the prospects connected with the application of the blockchain technology with other trending concepts like IoT, AI, and big data analysis can discover new possibilities and approaches to create value for the stakeholders in the maritime business domain. Finally, it is crucial to encourage collaboration between the stakeholders and participants in the industry, including the technology providers, regulators and academics to help identify barriers to adoption, dissemination of information and the promotion of innovation in the application of blockchain in the maritime sector. Despite the findings presented in this paper, it is important to note certain limitations that might affect the analysis of the results obtained here. Our research has delivered useful information but readers must recognize specific constraints within the data interpretation process. The required use of peer-reviewed journal articles restricted useful information found in industrial white papers and technical reports as well as conference proceedings where new technologies usually appear first [67]. The analysis confirmed the existence of a major conceptual-empirical divide in blockchain maritime research which matches the observation from Balci and Surucu-Balci [50] about minimal experimental data and theoretical dominance in this research field. The barriers related to adoption highlighted in the study by Yang [19] are in line with the challenges we have outlined above. This is especially most appropriate given industry-specific factors, like the maritime culture, not forgetting the complex multijurisdictionality of the business world. The standardization requirements as stated by Philipp et al. [43] need to be extended with maritime-specific standards for blockchain that address industry-specific maritime needs over generic blockchain protocols. A research agenda concerning legal and regulatory topics reacts to Kapidani et al.'s [57] observation about regulatory uncertainty being among the main obstacles for blockchain implementation in maritime developing economies. The outcomes derived from this investigation possess substantial importance to every stakeholder operating in the maritime domain. The implementation process for port authorities and terminal operators should treat blockchain as a combination of technological development with social and technical organizational management [21]. The research clusters show technology vendors what specific maritime blockchain solutions to develop which would include specialized blockchain technology solutions for both container tracking and document digitization since these use cases present strong market potential [46]. Specifically, the policymakers should aim at presenting future regulation policies that align this unique feature of blockchain technology with the maritime laws and data protection acts [49]. The following are some of the crucial areas that should be investigated by academics in great detail: Based on our findings, the following strategies are specifically recommended to enhance blockchain research and its implementation in the maritime sector. First, it is essential to pilot investigations into the effectiveness of blockchain using methodologically sound case studies. This approach supports the empirical advancement of research and helps establish the real-world applicability of blockchain technology. Secondly, the agenda emphasizes the need to advance the development of blockchain maturity models tailored to various domains, with a particular focus on the maritime industry. These frameworks would provide benchmarks for assessing the evolution and utilization of blockchain-related initiatives in this specialized sector. Thirdly, critical attention should be given to analyzing governance structures and institutions within multi-stakeholder blockchain systems and networks. A deeper understanding of these elements is crucial for fostering synergy, trust, and compliance among stakeholders with diverse backgrounds. Finally, the agenda encourages research on non-functional requirements and the exploration of new paradigms for integrating blockchain solutions with emerging technologies such as IoT, AI, and digital twin. In as much as this as an apparent, integrating such disciplines is crucial in the

development of smart port systems, especially as observed by other authors [10,47,36]. This approach seeks to align technology with the development of the overall processes of the maritime business with a view of leveraging synergies. The application of blockchain technology brings various noteworthy advantages toward maritime single windows alongside port community systems because it provides better security together with enhanced transparency and improved efficiency and built-in trust elements. The complete transformation of maritime single windows and port community systems depends on overcoming major technical barriers and organizational barriers as well as regulatory challenges through government-industry-academia collaboration. Blockchain technology functions best as part of an interconnected technology framework which addresses modern maritime operation complexities together with trading challenges [52,29,38]. Further research should explore the blockchain architectures designed to meet the precise requirements of MSW and PCS, while studying the legal and governance structures that enable secure blockchain adoption in maritime environments. In addition, to encourage studies that assess performance indicators such as cargo handling efficiency and customs clearance time together with data quality enhancement.

CRediT authorship contribution statement

Adil Nasser: Writing – original draft. **Fatima Ouzayd:** Writing – review & editing, Validation, Supervision, Methodology, Investigation. **Hamid Ech-cheikh:** Writing – review & editing, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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